

# Devisree Sumesh

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## Education

### Amrita Vishwa Vidyapeetham

B.Tech, Electrical and Computer Science Engineering / Minor: Automation & Robotics / CGPA: 8.58 2023 – 2027 (expected) Coimbatore, India

### Chavara Public School

Class 12, CBSE / 91.7% 2021 – 2023 Pala, India

### The Indian School

Class 10, CBSE / 97.4% 2009 – 2021 East Riffa, Bahrain

## Research

### AI in Robotic Surgery (BDF Royal Medical Services, Bahrain) [Live · GitHub]

2026 – Ongoing

- Conducting IRB-approved research applying deep learning to Hugo RAS robotic surgical video, supervised by Dr. Ayman Raees (Robotic Surgery Programme Director, BDF Hospital) and facilitated by Prof. Martin Corbally (RCSI Bahrain).
- Building a surgical instrument detection and skill-assessment pipeline on Hugo RAS video for downstream performance metrics.

### Hierarchical Safe RL with Control Barrier Functions for UAV Navigation [GitHub]

2026 – Ongoing

- Designing a hierarchical framework for single-UAV autonomous navigation combining Soft Actor-Critic path planning with a Control Barrier Function-QP safety layer for formal collision avoidance guarantees, plus an event-triggered replanning mechanism for computational efficiency.
- Implemented several custom obstacle-avoidance simulation environments in PyBullet for training and evaluation.

### $\mu$ PAD Liver Enzyme Spectral Analysis — AMBE Lab, Amrita [GitHub]

2025 – Ongoing

- Designed and implemented an automated computational calibration framework (AIC-based model selection, bootstrap-validated channel ranking, multi-channel sensor fusion) for a multiplexed optical point-of-care liver panel spanning AST, ALP, total protein, and bilirubin, using an AS7341 11-channel spectral sensor.

### Wearable Edge AI for Knee Rehabilitation Monitoring — First Author

2025 – 2026

Accepted, Student Track, IEEE ISTAS Kerala 2026

- Built an on-device Edge AI pipeline for knee flexion classification (5 angle classes, 95–100% on-device accuracy) on a NodeMCU ESP8266 with an MPU6050 IMU, quantised to int8 TFLite for real-time on-microcontroller inference.
- Integrated a browser-based 3D knee joint visualizer for remote clinical monitoring.

## Projects

### Teleoperated Surgical Robotic Arm with ROS 2 Motion Planning [Arm · ROS 2 Stack]

2025 – Ongoing

- Building a 4-DOF wireless teleoperated robotic arm using ESP32(ESP-NOW) and a PCA9685 servo controller with an ESP32-CAM wrist mount streaming live video to the detection model for a live surgical field view.
- Integrating detection output into a ROS 2 Jazzy / MoveIt2 motion-planning pipeline (URDF modelling, Gazebo simulation, OMPL RRTConnect) and mapping detected instrument labels to joint-space motion goals.

### Surgical Instrument Detection — YOLOv8 on Laparoscopic Video

2025 – 2026

- Trained YOLOv8n surgical instrument detectors on laparoscopic datasets (CholecSeg8k, m2cai16, CholecT50; 7 instrument classes, mAP50 up to 0.950), exported to ONNX for downstream inference.

### ORGuard — Full-Stack Surgical Safety Platform [Live · GitHub]

2026

- Built and deployed a full-stack MEN web platform digitising the WHO Surgical Safety Checklist; REST API with 12 endpoints including aggregation pipelines for compliance and phase drop-off analytics.

### Attention U-Net — Brain Tumour Segmentation [GitHub]

2025

- Trained Attention U-Net on the LGG MRI Dataset (110 patients, 3929 slices); Dice: 0.8388, IoU: 0.7270 on held-out test set.

### Surgical-VQA — LoRA Fine-Tuning of Qwen2-VL for Laparoscopic Scene Understanding [GitHub]

2026

- Fine-tuning Qwen2-VL-2B-Instruct 4-bit, (PEFT/LoRA) for visual question answering on laparoscopic cholecystectomy frames from CholecSeg8k; built a 40,400-pair VQA dataset across 13 anatomical/instrument classes.

## Skills

|           |                                                             |
|-----------|-------------------------------------------------------------|
| Languages | Python, C, SQL                                              |
| ML / DL   | PyTorch, TensorFlow, YOLOv8, OpenCV                         |
| Robotics  | ROS 2, Gazebo, MoveIt2, URDF/Xacro, TF2, RViz2, PyBullet    |
| Embedded  | ESP32, STM32 (Cortex-M4)                                    |
| Tools     | Git, Google Colab, VS Code, L <sup>A</sup> T <sub>E</sub> X |

## Extracurriculars

- Volunteer tutor for students (Free Tutoring Center) and for domestic violence survivors (STEM to Stability); STEM project builder for underserved girls (BuildHer); building the website for Child Survival Aid Ghana's Eco-Relief Foundation. (Ongoing)
- Special Interest Group Lead, Robotics and Automation under the EEE Department, Amrita Vishwa Vidyapeetham (2024)